
ME 501: Materials Science for Failure Analysis

Course Code	ME 501
Course Name	Materials Science for Failure Analysis
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	MS/Ph.D.
Mutual Exclusion	None
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Introductory remarks on Materials Science in the context of Engg:** Structure of perfect and imperfect solids; Elastic deformation and stress distribution, stress-strain relations under uniaxial loading.
- **Plastic Deformation in Crystalline Solids:** Introduction, theoretical strength of crystals and the motion of dislocation, energy of a dislocation and stable Burgers vectors. Slip planes and slip systems, relation between dislocation movement and plastic flow, dislocation generation, other modes of Deformation in crystalline solids.
- **Some Strengthening Mechanisms:** The phenomenon of yield point and strain hardening. Theories of yielding and strain hardening. Recovery, mechanisms of deformation at elevated temperatures, creep.
- **Mechanism of Fracture:** Ductile - Brittle transition, fracture. Design criteria for materials, environmental effects. Mechanical behavior of engineering materials under fatigue.
- Selection of materials and Processes, case studies.

Text and Reference Books

1. George E. Dieter, **Mechanical Metallurgy**, McGrawHill Book Company.
2. R. K. Honeycombe, **Plastic Deformation of Metals**, EWP
3. William D. Callister Jr., **Materials Science and Engineering**, Wiley India (P) Ltd.
4. Knott, **Fundamentals of Fracture Mechanics**.
5. A.H. Cottrell, **Mechanical Properties of Matter**, Wiley
6. Brown, **Introduction to Mechanical Behaviour of Materials**, Wiley.